

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (AIML / CCE) (Credit System) Degree Examinations

20AM611 / 20CC611 - CLOUD COMPUTING

May - June 2023

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	List hardware architectures for parallel processing. With neat diagram explain any two.	10	L*2	1	1
b)	Explain the two fundamental system architectural styles which cover the physical organization of components and processes over a distributed infrastructure.	10	L2	1	1
2. a)	Explain various software architectural styles of distributed computing.	10	L2	1	1
b)	Discuss RPC Reference model with neat diagram and explain how it enables inter-process communication.	10	L2	1	1
3. a)	List and explain the various models for message-based communication.	6	L2	2	1
b)	List the characteristics of Service Oriented computing.	5	L1	2	1
c)	Write a short note on following distributed object frameworks: i) Common object request broker architecture (CORBA) ii) Distributed component object model (DCOM/COM1) iii) Java remote method invocation (RMI)	9	L2	2	1
Unit – II					
4. a)	What is virtualization? List and explain characteristics of virtualization.	10	L2	3	2
b)	What are hypervisors? Compare Type 1 vs. Type 2 hypervisor with the help of neat diagram.	10	L2	3	1
5. a)	Discuss different theorems that define the properties that hardware instructions need to satisfy in order to efficiently support virtualization.	10	L2	3	1
b)	List and explain the pros and cons of virtualization.	10	L2	3	2
6. a)	Briefly explain security requirements for the architecture.	5	L2	4	1
b)	What are the security standards should include for securing cloud?	5	L2	4	2
c)	With a neat security framework diagram explain the multiple activities involved in effectively managing cloud security risk.	10	L2	4	2
Unit – III					
7. a)	With neat diagram, discuss the security monitoring and security feedback in cloud computing.	10	L2	5	1
b)	Discuss the purpose of cloud security monitoring.	5	L1	5	1
c)	List the reasons why security is ineffective from a software perspective.	5	L1	5	1
8. a)	Write short notes on Google App Engine Architecture.	10	L2	5	1
b)	With neat diagram, explain Microsoft Azure Architecture.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
